# Rachel Cooray

**Analyst, Data and AI, OCTAVE, John Keells Holdings**

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Analyst, Data and AI at OCTAVE, John Keells Holdings, applying AI and analytics to real business problems. I own four AI and analytics use cases for an FMCG distributor network and design how each one is evaluated in the field. Published researcher at IEEE ICAHS 2025, co-founder of Layer1 Studio, and First Class graduate in Computer Science from the University of Westminster.

## EXPERIENCE

**Analyst, Data and AI**, OCTAVE, John Keells Holdings (Colombo, hybrid) Jul 2026 to present

- Own four AI and analytics use cases on the distributor side of an FMCG business: distributor margin incentives, a replenishment system that recommends redistribution volumes, a permanent dispatch plan that minimises lorry costs, and a distributor efficiency model that increases line items per bill.
- Design how each use case is checked: pilot and control groups where possible, before and after where not. Start with operational measures such as billing activity, outlet reach and recommendation adherence, then margin. Report results without a control group as an association, not proven impact.
- Join regular governance calls with area, regional and sales managers and bring field learning back into each use case. Help prepare answers to challenging questions and slides that lead with the finding for leadership and finance.
- Keep reasoning on record through decision logs, process maps and problem statements. Reconcile numbers across dashboards and files and own the corrections. Tools: Excel, Databricks, Power BI.

**Analytics Delivery Associate**, OCTAVE, John Keells Holdings (Colombo) Apr 2026 to Jul 2026

**Data Science and Analytics Delivery Intern**, OCTAVE, John Keells Holdings (Colombo) Oct 2025 to Apr 2026

- Segmented distributors by performance and potential and helped design KPI-based incentive structures for an FMCG distributor network.
- Ran margin and sales-incentive scenario analysis and built Power BI dashboards for management to track targets and distributor performance.

**Co-founder and Developer**, Layer1 Studio (Part-time, remote) Dec 2025 to present

- I co-founded Layer1 Studio and build web and mobile products for small businesses, from shaping a brief to shipping and maintaining it.
- Web and mobile applications in Flutter, React and Node.js, from UI design to backend integration and deployment.

**Data Engineering Intern**, London Success Academy (Remote, United Kingdom) Sep 2025 to Oct 2025

- Built ETL pipelines on GCP for clickstream and transaction data, scaled processing with PySpark and Parquet partitioning, and produced revenue, daily active user and conversion funnel insights.

**Earlier internships at OCTAVE** 2022 and 2023

Analytics Delivery Intern, Mar 2023 to May 2023. Data Science Intern, Mar 2022 to Jun 2022.

## RESEARCH

**PCOS Care: A Machine Learning-Based Web Application for Early Risk Prediction of Polycystic Ovary Syndrome**, IEEE ICAHS 2025, Tunisia 2025

- DOI 10.1109/ICAHS66950.2025.11553077. With Malarvizhi Kaniappan Chinnathai and Salma Chahed, University of Westminster.
- Third place of more than 400 final year research projects, University of Westminster, 2025.
- An extended paper, "A Machine Learning-Driven Framework for Early Risk Stratification of Polycystic Ovary Syndrome", is under review for the MSIT Journal special edition.
- Deployed app: pcos-care-web-app.onrender.com. ORCID 0009-0009-3192-2669.

## EDUCATION

**BSc (Hons) Computer Science, First Class**, University of Westminster, United Kingdom Sep 2022 to Jul 2025

Specialised in Data Science. Started at IIT Sri Lanka and completed the degree at Westminster.

**CIMA Certificate in Business Accounting**, Chartered Institute of Management Accountants, United Kingdom Aug 2024

Economics, management and financial accounting, business law and corporate governance.

**St. Bridget's Convent, Colombo**, A-Levels in Mathematics, Physics and Chemistry 2008 to 2022

Head Prefect, 2020.

## SELECTED PROJECTS

**Finance education study assistant**, Full stack developer and AI engineer

A study assistant for a professional finance qualification that answers only from the syllabus, using a custom retrieval pipeline on Gemini, with two study modes and subject locking. React, Node.js, Gemini API, LangChain, Tailwind CSS.

**GenAI chatbot for neurodiversity support**, AI engineer, Microsoft EMBRACE hackathon finalist, 2024

A chatbot prototype that guides people newly diagnosed with neurodiverse conditions, and their carers, through what comes next. Built with a small cross-functional team under hackathon time limits. Generative AI, Conversational design, Prototyping.

**Carbon Footprint Tracker**, Data analyst and UI designer

Turns everyday activities into an estimated carbon footprint using activity-based emission factors, with interactive dashboards and suggestions for cutting it. Python, Flutter, Serverpod, Data visualisation.

**CV reviewer assistant**, Frontend developer and AI engineer

Recruiter-style CV feedback from Gemini, with client-side PDF processing. React, Gemini API.

**Futsal team recommender**, Mobile app developer

A Flutter app that matches players to teams using collaborative and content-based filtering, with centralised stadium booking. Flutter, Node, Flask, Machine learning.

**CSR street-vendor analytics**, Field researcher and analyst

A field survey of street food vendors whose findings the CSR team adopted. Data collection, Data analytics, Field research.

**Fantasy Coin Collector**, Game developer

A 2D dungeon crawler built from scratch in HTML5 Canvas, with its own physics, procedural maps and sound. HTML5 Canvas, JavaScript, Web Audio API.

## SKILLS

**Analytics and AI:** Machine learning, Statistical testing, Experiment and pilot design, Retrieval-augmented generation, LLM application design

**Languages and libraries:** Python, SQL, Pandas and NumPy, Scikit-learn, PySpark, JavaScript and TypeScript, Dart

**Platforms and tools:** Databricks, Power BI, Excel, Google Cloud Platform, Git and GitHub

**Web and mobile:** React and Next.js, Node.js, Flutter, Flask

**Working with the business:** Problem statements and decision logs, Pilot and control evaluation, Presenting to leadership and finance, Financial accounting

## TEACHING AND OTHER ROLES

**Class Teacher**, St. Mary's Church, Dehiwala Aug 2022 to Aug 2024

**Student Support Worker**, Clear Links Support Oct 2024 to Apr 2025

**Customer-facing roles while studying in the UK**

Customer Advisor, UNIQLO, Aug 2025 to Oct 2025. Customer Service Associate, EG On The Move, Jan 2025 to Sep 2025. Retail Associate, TK Maxx, Nov 2024 to Jan 2025.

## ACHIEVEMENTS

**Third place, Final Year Research Project**, University of Westminster, judged with Netcompany and SmartDCC 2025

**Hackathon finalist, GenAI for neurodiversity**, Microsoft EMBRACE Employee Resource Group 2024

**New Generation Awards**, Women in Management and the National Youth Services Council 2020